

Java-Based GUI Sales Management System

Technical Report

1. Introduction

This project presents the design and implementation of a Java-based Sales Management System with a Graphical User Interface (GUI) and database integration. The system simulates a small retail business environment where products are sold to customers, and transactions are recorded systematically.

The application is developed using Java Swing for the GUI and MySQL as the database backend. It demonstrates the use of Object-Oriented Programming (OOP) principles, database connectivity using JDBC, and basic software testing.

2. System Objectives

The main objectives of this system are:

- To provide a user-friendly interface for managing sales operations
- To manage product information (add, update, delete, view)
- To manage customer details
- To process sales transactions efficiently
- To automatically calculate totals and generate invoices
- To store and retrieve data using a relational database
- To enhance understanding of OOP and database integration

3. Requirements Analysis

3.1 Functional Requirements

The system supports the following functionalities:

- User authentication through a login screen
- Product management:
 - Add new products
 - Update product details
 - Delete products
 - View product list

- Customer management:
 - Add customers
 - View customer list
- Sales processing:
 - Select customer
 - Select products and quantities
 - Validate stock availability
 - Calculate subtotal and total
 - Store sales in database
- Invoice generation:
 - Display invoice
 - Export invoice as PDF

3.2 Non-Functional Requirements

- User-friendly GUI using Java Swing
- Input validation (no empty or invalid values)
- Error handling using try-catch blocks
- Data persistence using MySQL database
- Clean and maintainable code structure

4. GUI Screenshots

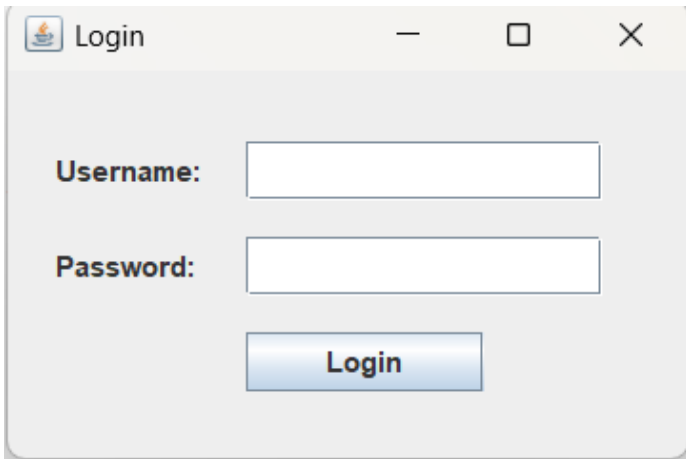


Figure 1. Login Screen

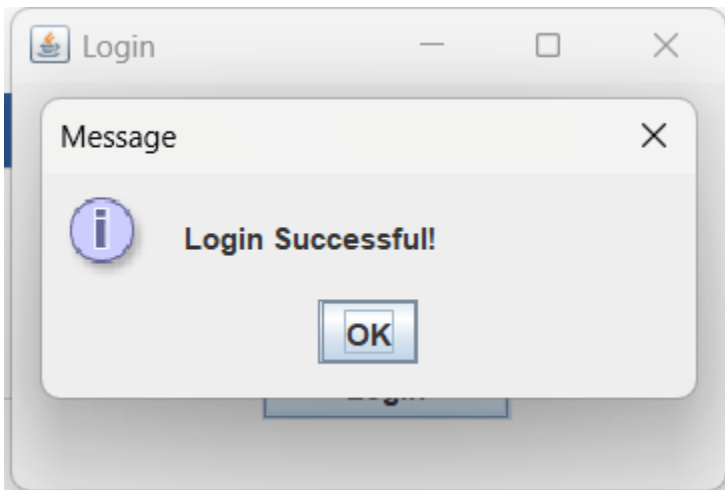


Figure 2. Login Success Message

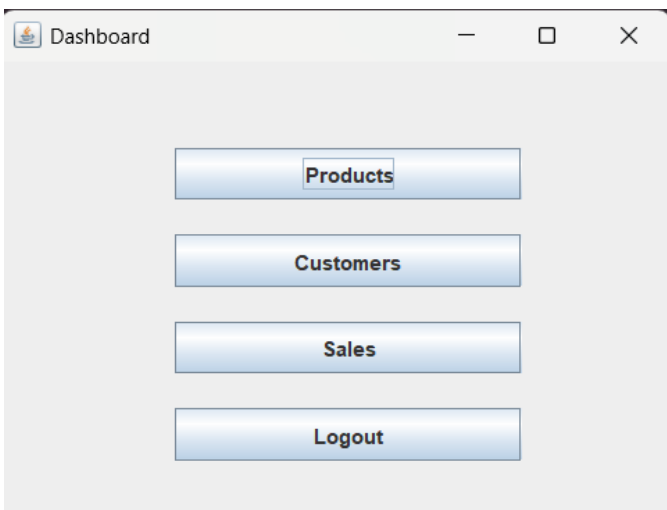


Figure 3. Dashboard

Product Management Screen

Product Management

Name: Add

Price: Update

Quantity: Delete

ID	Name	Price	Qty
1	pen	30.0	400
2	Book	100.0	400
3	Eraser	15.0	400
4	Sharpner	25.0	400
5	Pencil	30.0	400
6	Scissor	150.0	400
8	Marker	100.0	400

Figure 4. Product table screen

Product Management

Name: A4 Add

Price: 5 Update

Quantity: 1000 Delete

ID	Name	Price	Qty
1	pen	30.0	400
2	Book	100.0	400
3	Eraser	15.0	400
4	Sharpner	25.0	400
5	Pencil	30.0	400
6	Scissor	150.0	400
8	Marker	100.0	400

Figure 4. Adding a product

Product Management

Name: A4 Add

Price: 5 Update

Quantity: 1000 Delete

ID	Name	Price	Qty
1	pen	30.0	400
2	Book	100.0	400
3	Eraser	15.0	400
4	Sharpner	25.0	400
5	Pencil	30.0	400
6	Scissor	150.0	400
8	Marker	100.0	400
9	A4	5.0	1000

Figure 5. Updated product table after adding a product to stock

Customer Management Screen

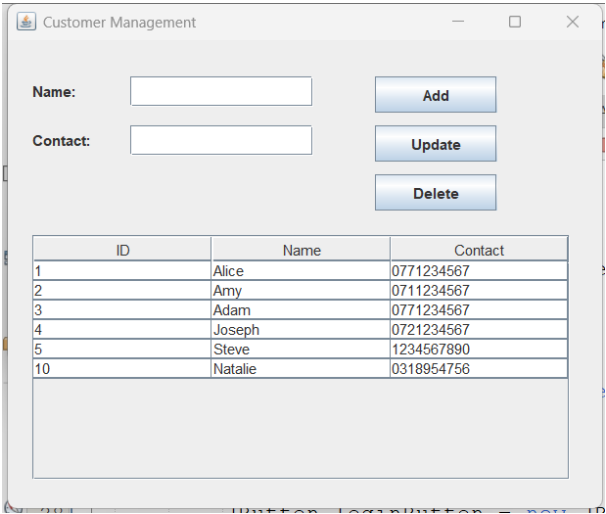


Figure 6. Customer table screen

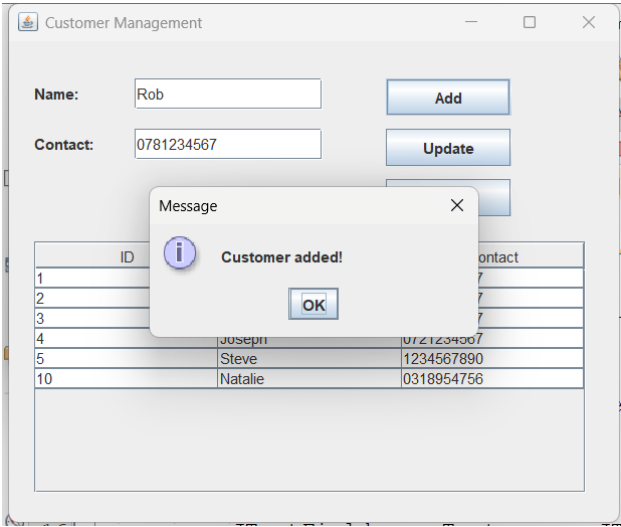


Figure 7. New customer added to the table

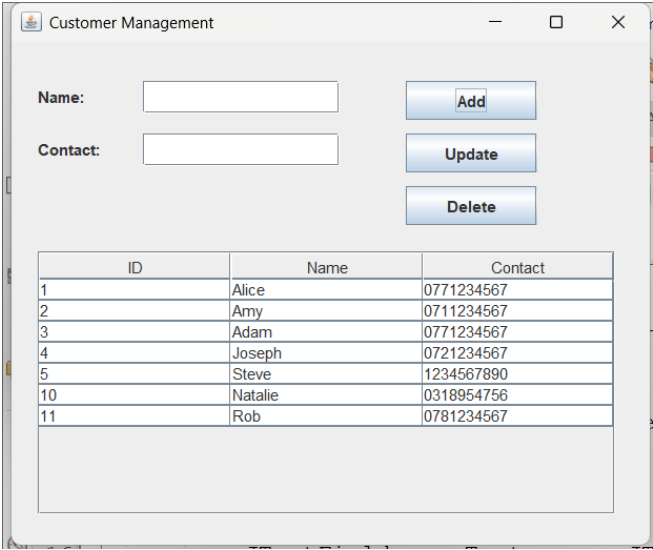


Figure 8. Updated customer table after adding a new customer

Sales Screen

The Sales screen window has a title bar with a standard icon and window controls. It contains a Customer dropdown menu set to "1 - Alice". Below it are input fields for Product ID and Quantity, both currently empty. To the right of the Product ID field is an "Add Item" button. Below the input fields is a table listing products:

ID	Name	Price	Stock
1	pen	30.0	400
2	Book	100.0	400
3	Eraser	15.0	400
4	Sharpner	25.0	400
5	Pencil	30.0	400

Below the product list is a summary table with columns: Product, Price, Qty, and Subtotal. It is currently empty. At the bottom left, it says "Total: 0". At the bottom center is a "Complete" button.

Figure 9. Sales screen

The Sales screen window now shows a new transaction. The Customer dropdown remains "1 - Alice". The Product ID field now contains "5" and the Quantity field contains "5". The "Add Item" button is still present. The product list table is the same as in Figure 9. The summary table now contains one row:

Product	Price	Qty	Subtotal
Pencil	30.0	5	150.0

The "Total: 150.0" is displayed at the bottom left, and the "Complete" button is at the bottom center.

Figure 10. Addin a new sale transaction

The Sales screen window is partially obscured by a "Message" dialog box. The dialog box has a title bar and an information icon. It displays the following text:

===== SALES INVOICE =====
Date: 2026-03-30T12:05:19.611302700

Customer ID: 1
Customer Name: Alice

Product Details:

Product ID: 5
Product Name: Pencil
Price: 30.0
Quantity: 5
Subtotal: 150.0

TOTAL AMOUNT: 150.0
=====

At the bottom of the dialog box is an "OK" button. In the background, the Sales screen shows the same state as Figure 10, with the "Total: 150.0" and "Complete" button visible.

Figure 11. Invoice generated after a sale

Product Management

Name: Add

Price: Update

Quantity: Delete

ID	Name	Price	Qty
1	pen	30.0	400
2	Book	100.0	400
3	Eraser	15.0	400
4	Sharpner	25.0	400
5	Pencil	30.0	395
6	Scissor	150.0	400
8	Marker	100.0	400
9	A4	5.0	1000

Figure 12. Reduced stock of product table after a sale

===== SALES INVOICE =====

Date: 2026-03-30T12:05:19.611302700

Customer ID: 1

Customer Name: Alice

Product Details:

Product ID: 5

Product Name: Pencil

Price: 30.0

Quantity: 5

Subtotal: 150.0

TOTAL AMOUNT: 150.0

=====

Figure 13. Generated invoice after a sale

5. Database Design

The system uses a MySQL database named salesdb.

Tables Description:

Customer Table

- customerId (Primary Key)
- name
- contact

Product Table

- productId (Primary Key)
- name
- price
- quantity

Sale Table

- saleId (Primary Key)
- customerId (Foreign Key)
- total

SaleItem Table

- id (Primary Key)
- saleId (Foreign Key)
- productId (Foreign Key)
- quantity
- subtotal

The database is connected using JDBC through a dedicated connection class.

6. Class Diagram

The system is designed using Object-Oriented Programming principles.

Core Classes:

- Product
- Customer
- Sale
- SaleItem

Data Access Classes (DAO):

- ProductDAO
- CustomerDAO
- SaleDAO

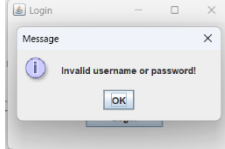
User Interface Classes:

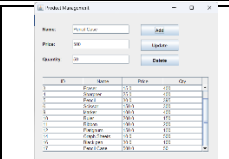
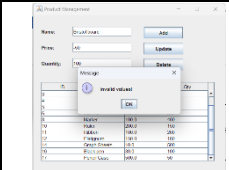
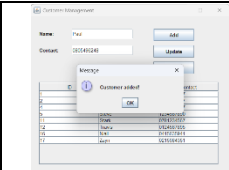
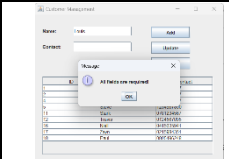
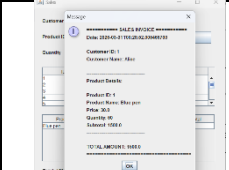
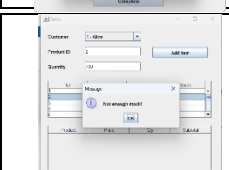
- LoginUI
- DashboardUI
- ProductUI
- CustomerUI
- SalesUI

The system follows a layered structure where UI, business logic, and database operations are separated.

7. Testing and Test Cases

The system was tested using functional testing methods.

Test Case	Input	Expected Output	Result	
Login valid	Correct username & password	Dashboard opens	Pass	
Login invalid	Incorrect credentials	Error message	Pass	

Test Case	Input	Expected Output	Result	
Add product	Valid data	Product added to database	Pass	
Invalid product input	Negative price/quantity	Error message	Pass	
Add customer	Valid details	Customer added	Pass	
Empty fields	Blank input	Validation message	Pass	
Create sale	Valid data	Sale recorded and invoice generated	Pass	
Stock validation	Quantity exceeds stock	Error message	Pass	

8. Challenges and Limitations

Challenges:

- Implementing database connectivity using JDBC
- Designing user-friendly GUI layouts using Java Swing
- Handling user input validation and error messages
- Managing sales transactions and updating stock correctly

Limitations:

- Login system uses hardcoded credentials
- No role-based access control
- Limited reporting features
- Basic GUI design without advanced styling

9. Conclusion

The Sales Management System was successfully developed using Java, Swing, and MySQL. The system meets all functional requirements including product and customer management, sales processing, and invoice generation with PDF export.

The project demonstrates practical implementation of OOP concepts, GUI development, and database integration. Future improvements can include enhanced security, advanced reporting, and improved user interface design.

Figure 1. Login Screen 3

Figure 2. Login Success Message 3

Figure 3. Dashboard 3

Figure 4. Dashboard**Error! Bookmark not defined.**

Figure 5. Adding a product..... 4

Figure 6. Updated product table after adding a product to stock 4

Figure 7. Customer table screen 5

Figure 8. New customer added to the table 5

Figure 9. Updated customer table after adding a new customer 5

Figure 10. Sales screen..... 6

Figure 11. Addin a new sale transaction 6

Figure 12. Invoice generated after a sale 6

Figure 13. Reduced stock of product table after a sale 7

Figure 14. Generated invoice after a sale 7